

[MPL-19003] GEOINFORMATICS	
Teaching Scheme:	Examination Scheme:
Lectures: 3 Hrs/ week	T1 and T2 - 20 marks each
	End Sem. Exam. - 60 marks

Course Outcomes: At the end of the course, the students is able to
CO_1: Explain and Describe the components and functionality of a GIS.
CO_2: Understand the nature of geographic information and conduct simple spatial analysis.
CO_3: Explore mapped data by relating GIS with remote sensing technologies.
CO_4: Learn to use GIS software used for comprehensive planning, zoning, land use inventories, site suitability assessments, and socio-demographic analysis.

Unit 1:	Remote Sensing	[8 Hrs]
	Introduction to Remote Sensing – Aerial photography, photogrammetry, Definition of Remote sensing, A tool for resource surveys, Application in land use, land cover analysis and town planning.	
	Electromagnetic Energy – Definition, Properties velocity, wavelength, frequency. Electromagnetic spectrum: definition, wavelength regions, Imaging system, satellite images.	
Unit 2:	Aerial Photogrammetry	[8 Hrs]
	Aerial photography – Types of aerial cameras, Types of photographs, vertical, horizontal, oblique. Geometry of Aerial photographs – Tilt, Swing, Photomadir, Principle points, principal distance, flying height, air base, overlaps, sidelaps, methods of scale determination, Image displacement due to relief and tilt, stereoscopic vision.	
	Satellite Imageries – Satellite platforms and orbit patterns, optical, mechanical, scanners, Infrared scanners, earth resource technology satellite, LANDSAT, SPOT, and IRS.	
Unit 3:	GIS	[8 Hrs]
	Introduction – Definition and meaning, Application to town planning. GIS Vs. Maps – Advantages and disadvantages	
	Data Modes for GIS – Raster based data – Rasterization using toposheets, isolines maps and urban maps. Nature of remote sensing data, resolution.	
	Vector based data – Digitization of point, line aerial boundaries using graphs	
Unit 4:	GIS Component	[8 Hrs]
	Components of GIS – (1) Input (2) Preprocessing (3) Spatial data base – point, line, polygon (4) Analysis (5) Output	

	Introduction to one Raster based and one vector based GIS package. Application of special functions and compatibility for various types of database.
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Reference Books:

1. Interpretation of Aerial Photography by Thomas Eugene Avery: Bungs Pub. Co., Minnesota.
2. Remote Sensing, Principles and Applications by Floyd Sabins; Freeman and Co, New York.
3. Principles of Remote Sensing by P. J. Carran : (ELBS).
4. Remote Sensing and Image Interpretation by Lillesand Thomas M. & Kiefer Ralph W: (John Wiley & Sons Inc. New York)
5. Geographical Information System by D. Hall & R. Babbage (1989) :
6. Bihar – GIS by Lahiri M. (1993)

- Give the mapping of COs with POs and PSOs - indicating dash- no mapping, 1- Low, 2 - Med, 3 - High

POs & PSOs COs	PO1	2	3	4	5	6	PSO1	PSO2	PSO3
CO1	-	-	1	-	-	1	3	2	2
CO2	-	-	1	-	2	1	3	1	-
CO3	-	-	3	-	1	1	-	2	1
CO4	-	-	3	-	2	-	3	3	-